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# Artificial Intelligence and Deep Learning-based Model for Indoor Environment: with Virtual Reference Tag Allocation

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**Abstract**

By the evolution of context-aware application, indoor location positioning gains much more attention worldwide. Radio Frequency Identification (RFID) is one such wireless positioning technology adopted widely. Moreover, enhancing the tracking accuracy and thus reducing the tracking error remains a challenging task in RFID based indoor environment. So as to overcome this aspect, a localization and tracking method depending on virtual reference tag (VRT) is employed in the indoor environment. This research work includes four modules like (i) data collection and Received Signal Strength Indication (RSSI) estimation;(ii) Optimal allocation of VRT using Spider Monkey Optimization (SMO) at which the allocation of VRT is made by considering number of tags, environmental factors (humidity and temperature), and SNR. Based on this data, the RFID reader allocates VRT for each grid so as to increase the tracking accuracy; (iii) Artificial Neural Network (ANN) module-based localization at which the localization of unknown tags is made; and (iv) tracking of moving target tag is carried based on Aquila Swarm Optimization-Long short-term memory (ASO-LSTM) approach. The optimization at this stage is employed to choose optimal position which enhances the accuracy of tracking. Thus, finally the location estimation is carried by this approach. The simulation is carried in NS3.26 network simulator and the performance are estimated in terms of SNR, tracking error, mean error, tracking accuracy, mean time, and standard deviation. The performance value is then compared with traditional models to show the enhancement of proposed mode over the other conventional techniques.

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Keywords: Radio Frequency Identification (RFID), Received Signal Strength Indication (RSSI), localization, tracking, moving target, virtual reference tag, Spider Monkey Optimization (SMO), Artificial Neural Network (ANN), Aquila Swarm Optimization-Long short-term memory (ASO-LSTM), tracking error, tracking accuracy.

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## 1. Introduction

The emergence of location-dependent services and applications lead to rising demand of anywhere localization [1]. Though, the space-based satellite navigation systems like Global Positioning System (GPS) offers higher accuracy on localization in outdoor, the poor connection among the end-device and satellites makes them not accessible for indoor localization [2]. Several approaches on indoor localization were proposed thus far with the use of initial measurement unit (IMU), ultrasonic, FM, magnetic field, Radio-frequency ID tags (RFID), Wi-Fi and so on. Among various branches of methods, RFID that depends on received signal strength indicator (RSSI) have attracted the attention mostly in last decades [3]. The technology of existing GPS is not appropriate for tracking the indoor system [4] as it has the issue of signal blocking and the non-line of sight conditions. For addressing this issue, RFID based tracking technique is developed. This is capable of supporting huge-scale indoor environment and thus performs well in the condition of non-line of sight. is not suitable for the indoor tracking system since, it suffers from the signal blocking and non-line of sight conditions. To address this bottleneck, Radio Frequency Identification (RFID) based tracking technology is evolved. It supports a large-scale indoor environment and also performs well in non-line of sight conditions.

RFID localization method is capable of exchanging the data over a non-contact two-way communication for the reason of identification and positioning of the mobile devices [5]. The present usage of RFID technology for building the indoor positioning system has a great benefit. By the evolution of RFID software and hardware technology, the cost of production for the RFID tags were greatly reduced to few cents and an algorithm for the identification and allocation of multi-tag is becoming mature highly. The RFID readers performance is likewise superior [6]. This could read huge number of tags more quickly which is efficient extremely. The tracking system based on RFID technology consists of RFID tags and the readers. In this, the RFID readers are employed for collecting the RSSI information's from the moving target tag over multiple tags in the indoor framework .

A Deep-learning based method is employed for tracking moving objects in the environment of indoor tracking in this approach. In the traditional models, a range-free indoor mobile object tracking is employed using RFID system. The multi-direction weight position Kalman filtering based indoor tracking approach is employed for tracking the moving objects in indoor environment [7]. Though, these techniques possess high tracking error that affects the process of effective tracking and localization of the target tags. Specifically, there exists some demand for the accurate localization of moving target tags in the indoor environment which makes the need for the development and enhancement of advanced methodologies. Thus, the deep learning-based model is proposed in this research work for addressing the issues of existing RFID tracking system methodologies.the major intention of this paper is to investigate the usage of active RFID technology for the localization of movable objects equipped with RFID tags using virtual reference Tag allocation (VRTs) by employing LSTM algorithm.

The remaining part of the paper is organized as shown: section II is provided with a short and precise review on various traditional method employed based on the tracking and localization of RFID system. The proposed system methodology is narrated briefly in the section III. the performance assessment carried is projected in section IV. The entire manuscript is concluded in section V.

## 2. RELATED WORKS

A detailed survey was conducted on several research paper and its review were discussed below:

The author in the work [8] suggested a novel open-source RFID simulator which allows the modeling of environment and thus testing the RFID tags deployment and antennas apriori. The implementation of simulator is carried as Gazebo plugin, the mostly widely employed robotic framework that was integrated with the Robot operating system (ROS) so as to reach broad audience. For validating the simulator, modeling of warehouse scenario was considered, and an algorithm of tag localization which employs phase unwrapping method and intersection of hyperbolae technique employing a mounter reader antenna on mobile robot for estimating the tags position was deployed in this case.

This paper [9] proposes a method for RSSI based indoor localization and tracking in cluttered environments using Deep Neural Networks. We implemented a real-time system to localize people using wearable active RF tags and RF receivers fixed in an industrial environment with high RF noise. The proposed solution is advantageous in

analyzing RSSI data in cluttered-indoor environments with the presence of human body attenuation, signal distortion, and environmental noise. This method was successful in predicting the walking direction of the person.

A framework of generic asset tracking service was presented in [10] for the indoor environment. A comprehensive evaluation of existing approaches was introduced with the use of technology enablers evaluation matrix, that tries to identify the suitability of asset tracking models for the indoor location services. As a pilot testing, the seamless location capabilities emerged by HERE technology was estimated and a smart manufacturing demonstration center (SMDC). The asset tracking of this works both outdoor and indoor seamlessly and employes a Wi-Fi and GPS based models along with Bluetooth based tracking framework. The testing estimates the real-time seamless indoor environment tracking and aids in identifying the appropriate tracking model depending on the needs.

This paper [11] employs a particle swarm optimization (PSO) algorithm along with RSSI channel model for the target localization and tracking of indoor environment. The eight various models' performance combinations of regular or random pints, adaptive or fixed weights, and a region segmentation model (RSM) was proposed for the target tracking and localization process were investigated. The simulation outcome shows that the RSM model could reduce number of particles employed in PSO approach and thus enhances the tracking and positioning speed without affecting the target localization and tracking accuracy rate.

An indoor real-time location tracking model was presented in with the use of ultra-high frequency identification technology. A K-Nearest neighbor dynamic technology driven hidden Markov model (HMM)n was developed for tracking subsequent moving object positions on combining the static with dynamic localization tracking. The validation was carried statistically by the regression models for relating the RSSI tags indicator and distance, that were updated iteratively over a time for accommodating the changes in environment. The likelihoods of probabilistic tags' latent locations depend on the present location and the RSSI are employed for constructing the Markov model's emission and transition matrices, correspondingly[12].

A state-of-the-art model on UHF RFID localization was provided in [13] which offers a introduction on methods and applications. The majority of methods of UHF-RFID localization have two kinds like range-free and range-based which were reported in existing models. The range-based method attains angle or distance among the tag and antenna by utilizing RSSI, single phase, time information or tag location estimation on combing the geometrical characteristics scheme. A technique depending on the non-fingerprinting ad fingerprinting were discussed in range-free domain. At last, the comprehensive evaluation of the survey works was carried and conclusion with future trends were presented.

A review on traditional machine-learning based models for the IoT-dependent positioning system was presented in [14]. The existing models includes several approaches on distance localization and estimation;however, this approach have some limitations. Due to high prediction performance, ML techniques were employed for the addressing the indoor localization issues recently.

A LANDMARC location-based algorithm was presented in [15] which was employed in the regional location at the online phase, and ELM having VRT was introduced in locked regions. The scheme of indoor positioning system designed was based on Intel R1000 platform, and the system was thus realized by c++ language. The evaluation made shows that the average error was about 38% over the LANDMARC approach and was 19% for ELM approach. The system could enhance the indoor localization accuracy effectively in the low tag density regions.

A signal loss dependent approach-based RFID system was presented in [16]for localizing the moving objects. In this, filter was employed for the ambiguity of information gathered. Then, the triangulation approach was employed for localizing the moving objects position. The moving objects localization was attained over mapping the attained RSSI values. Depending on the mapping outcome, this in turn localizes the moving objects in an effectual manner.

The author in the work [17] presented a model of moving object localization with the use of RFID tag array information. This in turn uses specific filter for tracking the indoor environment's moving objects. Also, it employs laser ranging model for tracking the position of moving object. In this, the information regarding laser velocity were clustered with the use of density based spatial clustering of application with noise (DBSCAN) approach.

A novel localization scheme was presented in [18] that aims at existing RFID localization technology defects in the localization accuracy and the cost of deployment termed ANTspin: RFID efficient absolute localization model over spinning antenna, that introduces rotary table in an experiment. The reader antenna might be fixed to the rotary table for collecting dynamic data continuously. On comparing with static acquisition, there exists several

information on localization. Then, the relative incidence angle and distance among the antenna and tags could be analyzed for the purpose of localization with the characteristics of RSSI data.

A differential RSSI based tracking of moving object was introduced in the work [19]. This work considers the indoor environment at which the equipment utilized for the constructions were tracked with the use of RFID system. This in turn deploys four RFID readers or the purpose of tracking moving objects. This employs RSSI from the mobile target tag at various positions for tracking objects.

## 3. PROPOSED WORK

A narration of proposed workflow model is described in this section. The schematic view of the proposed structure is given in figure 1.

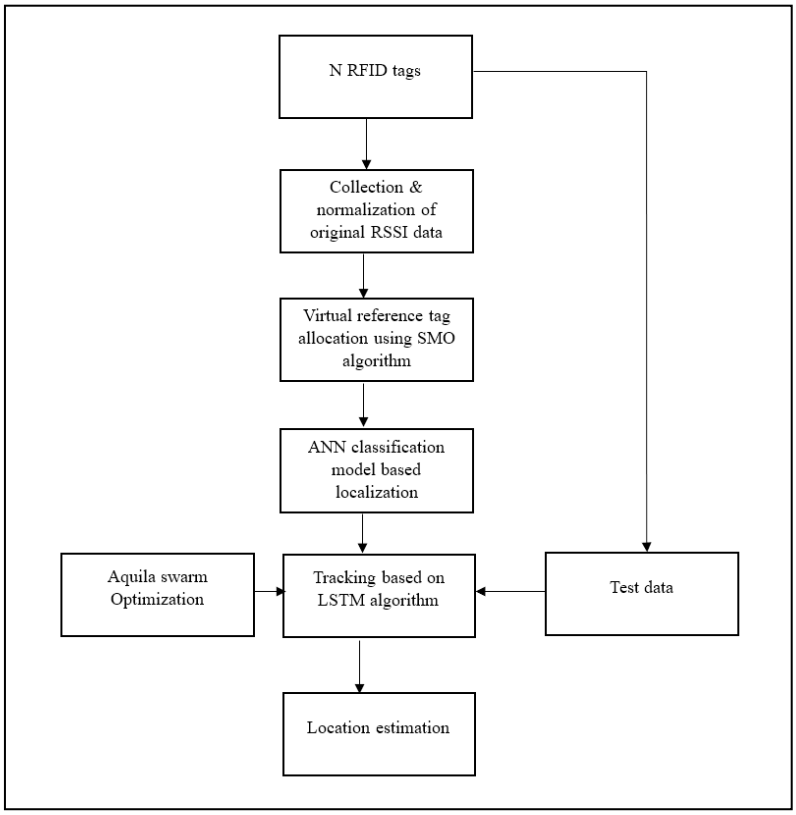



Figure 1 schematic view of proposed structure

*a. Data collection and RSSI estimation*

RFID reader initiates its communication with the tag over the half duplex communication mode of reader first speech. Thus, RFID communication involves two links like forward and backward link. In the RFID communication system, the reader antenna is responsible for transmitting the interrogation signals over the surrounding passive tags. Once the transmitted signal is received, tag will activate themselves with the energy signal and thereby backscatters signals to reader antenna. The RSSI is estimated from these signals through measurements. Practically, a complex environment might affect the wireless signal transmission. The algorithmic path loss model

was exploited for simulating the RFID signal propagation. RSSI is determined mathematically by the following equation:

$$RSSI[dbm] = P_t[dbm] + G_t[dbi] + G_r[dbi] - P_L(f,d)[dB] + X_\sigma \quad (1)$$

Here,

$$P_L(f,d) = 10nlog_{10}\left(\frac{4\pi f}{c}\right) + 10nlog_{10}(2d) \quad (2)$$

From equation 1, $P_t$ signifies the transmission signal power from the RFID reader, $G_t$ *and* $G_r$ signifies RFID reader transmitter and receiver antenna gains. $P_L(f,d)$ from equation 2 refers to the path loss with f as carrier frequency and a distance among tag and RFID reader is d, which is indicated by unit dB. The speed of light is represented by c. The path loss constant is denoted by n, which indicates the path loss rate increasing with distance and it is based on surrounding environment. $X_\sigma \sim \aleph(0,\sigma_1^2)$ signifies the gaussian distributed noise having zero mean and variance $\sigma_1^2$.

*b. Optimal allocation of virtual reference tag (VRT) using SMO*

In the tracking system, so as to overcome the issue of real reference tag in the indoor tracking, this work employs virtual reference tags in the indoor tracking. In the network of 3D grid, one real reference tag is placed at each grid. For reducing the interference imposed by the real reference tag. It is attained by reducing the real reference tag number through the virtual reference tag allocation. Hence, the virtual reference tags are allocated in the indoor environment by the RFID reader nodes placed kept on ceiling. The optimal allocation of virtual reference tag is achieved by using spider monkey optimization algorithm. This in turn performs greatly on comparing other existing optimization models like PSO, GWO, and ACO by offering outcomes within four iterations. This considers the succeeding parameters for computing the fitness function like SNR, number of tags, environmental factors like humidity, and temperature which are termed collectively as attributes. As these two environmental factors affects, the RSSI signal in indoor environment, RFID allocated virtual reference tags based on this information for each grid it monitors. In the SMO, the optimization depends on the foraging behavior of spider monkeys which is influenced by the time and energy needed to search food, availability of food source, and the predator's presence or competition in the surrounding environment. Likewise, the virtual reference tags are allocated depending on the attributes. At first, the virtual reference tag position is plotted depending on the distance among reference tag and is formulated by:

$$V_t = \frac{R_t}{r^2} \quad (3)$$

In this, $V_t$ is denoted by virtual reference tag, and reference tag is signified by $R_t$, and the distance among reference tag in grid is indicated by $r^2$. Thus, the interference is also considered and is minimized for facilitating the effectual communication among tags and is formulated as:

$$S_{V_t} = S_{R_t} e^{-\lambda y} \quad (4)$$

In this, the signal intensity of virtual reference tag is denoted by $S_{V_t}$ . The initial signal intensity of the reference tag is denoted by $S_{R_t}$, $\lambda$ signifies signal coefficient, and y refers to distance factor correspondingly. From this, availability is G is denoted by

$$G = A\delta\rho k_s^4 e^{-\lambda y} \quad (5)$$

In this, A is signified as the surface grid area, connectivity factor is denoted by $\delta$, humidity is signified by $\rho$ and temperature is referred by k. in this, the random distribution of food source is considered as per SMO which denotes that the method places reference tags randomly and it is signified as follows.

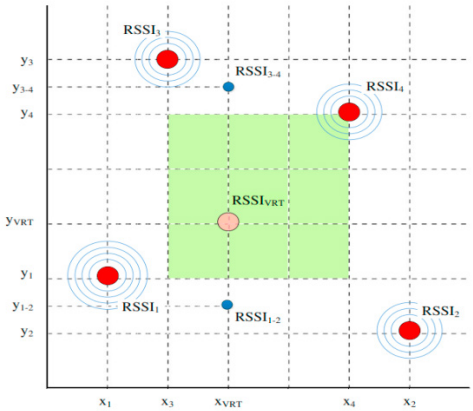



Figure 2 VRT allocation representations

By figure 2, for the case that the winner cluster covers four real reference tags, the linear interpolation at x-direction is carried as a first step. On employing equation 6, among $RSSI_1$ and $RSSI_2$outcomes in $RSSI_{1\text{-}2}$, and on employing equation 7 among $RSSI_3$ and $RSSI_4$ outcomes in $RSSI_{3\text{-}4}$, then, virtual reference tag RSSI could be attained by employing equation 8 which in turn interpolates at y-direction among $RSSI_{1\text{-}2}$ and $RSSI_{3\text{-}4}$.

$$RSSI_{1-2} = \frac{x_2 - x_{VRT}}{x_2 - x_1} RSSI_1 + \frac{x_{VRT} - x_1}{x_2 - x_1} RSSI_2 \tag{6}$$

$$RSSI_{3-4} = \frac{x_4 - x_{VRT}}{x_4 - x_3} RSSI_3 + \frac{x_{VRT} - x_3}{x_4 - x_3} RSSI_4 \tag{7}$$

$$RSSI_{VRT} = \frac{y_{3-4} - y_{VRT}}{y_{3-4} - y_{1-2}} RSSI_{1-2} + \frac{y_{VRT} - y_{1-2}}{y_{3-4} - y_{1-2}} RSSI_{3-4} \tag{8}$$

By this, the optimal allocation of VRT is carried by means of attaining fitness function from applied SMO optimization. The allocation done here depends on SMO foraging behavior.

*c. ANN module-based localization*

For tracking the moving target tag, the position estimation should be done initially using localization process. For this purpose, the technique of ANN is employed for estimating the exact position of moving target. For identifying the initial target tag position, RFID reader considers RSSI from the neighbor virtual reference tag. The existing methods employed for the localization process has some limitations like size of area, reference tag topology, and environment were not taken into consideration which does not offer best positioning for all environment. So as to overcome this issue, the proposed method employs ANN method as an alternative for positioning. The architecture of ANN is shown in figure 3 provided here.

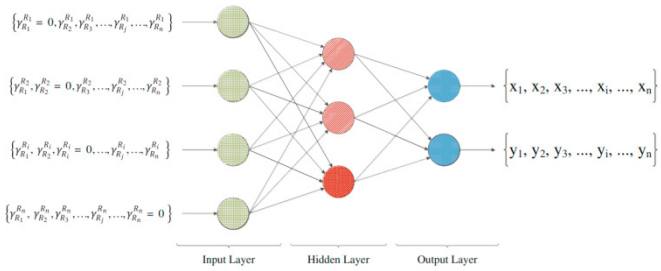



Figure 3 ANN architecture

In case, the network input is considered as n by the n matrix, then each row values are the dissimilarity of signal pattern among ith reference tag and each other reference tag. $\gamma_{R_i}^{R_j}$ refers to pattern dissimilarity or the distance indicator value among ithreference tag and jthreference tag (temporary target for training) after the collection of m data points. The matrix of $\gamma_R$will be constructed with the use of computed values from equation 9.

Apparently, each reference tag's signal dissimilarity with themselves will be equal to zero, hence the major diagonal entries are also zeros.

$$\gamma_{R_i}^{R_j} = \sqrt{\sum_{s=1}^{m}(RSS_s^{R_i} - RSS_s^{R_j})^2} \tag{9}$$

$$\gamma_R = \left\lfloor \gamma_{R_i}^{R_j} \right\rfloor_{i=1,\ldots n;\ j=1,\ldots n} \tag{10}$$

The hidden layer number is therefore set as one. In this work, the equation 11 is presented depending on the error and trial, which could offer the optimal number of hidden neurons layer on considering the input layer number and output layer nodes.

$$N_{hidden} = \left(\left(N_{input} + N_{output}\right) \times \frac{2}{3}\right) + 2 \tag{11}$$

For example, from the architecture of ANN provided, if $N_{input}$ is 19 which includes 4 real and 15 VRT and $N_{output}$ is equal to 2, the hidden neuron number is equivalent to sixteen.

*d. Tracking based on ASO-LSTM approach*

In the indoor environment, tracking of moving target is carried by means of Aquila swarm optimization-based LSTM approach. The presented model uses following parameters for tracking the moving target like phase, distance, previous co-ordinates, and orientation. On using this information in the tracking process, the proposed model tracks the moving target efficiently. The Aquila swarm optimization-based LSTM is employed in this work. The classifier is enhanced with the use of fitness function value attained from ASO. The technique of ASO is employed with a novel fitness function.

A distinctive ASO consist of 4 steps which the individual swarm may carry to capture the prey.

$Step$ 1: Expanded exploration

Initially, Aquila may recognize the prey location thereby widely fly in search of prey and is expressed as:

$$\mathrm{Y_i(t+1) = Y_{best}(t) \times \left(1 - \frac{t}{T}\right) + Y_M(t) - Y_{best}(t) * rand} \quad (12)$$

Here, $Y_i(t+1)$ signifies the individual position at iteration $t+1$. $Y_{best}(t)$ signifies the optimum location at present iteration. The position of whole individual mean at current iteration is signified by $Y_M(t)$.

$$Y_M(t) = \frac{1}{N}\sum_{i=1}^{N} Y_i(t) \quad (13)$$

Here, $Y_i(t)$ refers to $ith$ position of individual $at\ iteration\ t\&N$ signifies swarm's population size. $rand$ denotes the random number of Gaussian distributions among 0 and 1 interval.T is maximum allowed number of iterations.

$Step$ 2: Narrowed exploration

It is at the stage of exploration still, once Aquila identifies the prey, it might fly to the prey overhead on revolving & so makes the space & is signified as:

$$Y_i(t+1) = Y_{best}(t) \times Levy(D) + Y_R(t) + (y - x) * rand \quad (14)$$

Here, $D$signifies the problem-solving dimensionality. $Levy(D)$ Signifies Levy flights which might be estimated as:

$$Levy(D) = p \times \frac{\mu \times \sigma}{|\upsilon|^{\frac{1}{\beta}}} \quad (15)$$

Here, $p = 0.01$ refers to constant parameter , random numbers among 0 and 1 will bedesignatedas μ and ʊ . Likewise,σ is designated by:

$$\sigma = \frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\frac{\beta-1}{2}}} \quad (16)$$

Here, $\beta$ denotes a constant valuethat is fixed to 1.5. Γ refers to the gamma function.

A random candidate selected is indicated as $Y_R(t)$ at present iteration. The spiral shape is indicated as y and x& is assessed as:

$$y = r \times \cos(\theta) \quad (17)$$

$$x = r \times \sin(\theta) \quad (18)$$

$$r = r_1 + U \times D_1 \quad (19)$$

$$\theta = -\omega \times D_1 + \theta_1 \quad (20)$$

$$\theta_1 = \frac{3\pi}{2} \quad (21)$$

Here, $r_1$ denotesthe fixed number among 1 and 20. An integer from 1 to that of the problem length is meant by $D_1$ . A constant fixed number $\omega = 0.005$.

$Step$ 3: Expanded exploration

In the exploration process, Aquila may re-initialize a thing once they are unsuccessful in recognizing the prey, later they are responsible for position updating using following expression:

$$Y_i(t+1) = \alpha \times [Y_{best}(t) - Y_M(t)] + \delta \times [(UB - LB) \times rand + LB] \quad (22)$$

In this, $[UB - LB]$means the problems definitional domain given. About 2 fixed small numbersweredesignated as $\alpha$ $and$ $\delta$.

*Step 4:*Narrowed exploitation

As quickly the Aquila is nearer to prey, this may take the narrowed exploitation & is uttered as:

$$Y_i(t+1) = QF \times Y_{best}(t) - G_1 \times Y_i(t) \times rand - G_2 \times Levy\ (D) + rand \times G_1 \quad (23)$$

Here, QF signifies the employed quality function for searching the search space equilibrium & is assessedby:

$$QF = t^{\frac{2\times rand-1}{(1-T)^2}} \quad (24)$$

$$G_1 = 2 \times rand - 1 \quad (25)$$

$$G_2 = 2 \times \left(1 - \frac{t}{T}\right) \quad (26)$$

Thus by this, an optimal fitness function value is attained which enhances the detection rate and accuracy of classifier.

For the process of classification, LSTM method is employed for making decisions regarding the position of target to track them. It offers decision vectors and thus selects the candidate data. The vector range lies among 0 to 1 at which the LSTM omits vectors that are close to zero and thus preserves vector that are nearer to 1. The LSTM thus generates input gate I using present unit input $X_t$&previous hidden state of LSTM $H_{t-1}$at step $T$.

$$I_t = \sigma(W_I[H_{t-1}, X_t] + B_I) \quad (27)$$

$\sigma$ refers to activation function.

$W_I$weighted matrix

$B_I$is the bias between two connected modules.

The $f_t$ forget gate of LSTM having two input values as $H_{t-1}$ and $X_t$ is denoted by:

$$f_t = \sigma\left(W_f[H_{t-1}, X_t] + B_f\right) \quad (28)$$

Here, $W_f$ denotes the forget gates weight and $B_f$ refers to the bias matrices. $H_{t-1}$, $X_t$ were employed in $I_t$ and $I_t$ is desirable to adapt the cell state information $\widetilde{C_t}$ .

$$\widetilde{C_t} = \tanh(W_c[H_{t-1}, X_t] + B_c) \quad (29)$$

$C_t$ a current one will be related to preceding $C_{t-1}$& the input candidate $C_t$ will be intended by:

$$C_t = f_t * C_{t-1} + I_t * \widetilde{C_t} \quad (30)$$

At last, LSTM employs an output gate $O_t$ to identify the next hidden state $H_t$ that coversprevious information.

$$O_t = \sigma(W_O[H_{t-1}, X_t] + B_O) \quad (31)$$

$$H_t = O_t * \tanh(C_t) \quad (32)$$

$W_O$ – Weight of the output gates.
$B_O$ – Bias matrices of the output gate.

Thus, by this approach the tracking of moving target tag is identified more efficiently.

## 4. Performance Analysis

The estimation of performance is carried and the outcomes validated are projected here.

*a. Simulation environment*

A 3-D grid network topology is designed which covers RFID readers, one centralized server, 16 real/phy reference tags and 32 VRT and 1 target tag. The proposed model is experimented in 300×400 m simulation environment for the localization of RFID and tracking process with the use of AI techniques in the indoor environment.
The system specifications are as follows;
(a) Hardware specification: Hard disk-500 GB and RAM-4 GB with network simulator-NS3.26
(b) Software specification: OS-Ubuntu 14.04 LTS.
The simulation parameter is given in table 1 provided below:

Table 1 simulation parameter

| **Parameters** | **Values** |
|---|---|
| Simulation area | 300×400 m |
| Topology | 3D grid network |
| RFID reader | 4 |
| Centralized server | 1 |
| Real/Phy reference tag | 16 |
| Virtual reference tag | 32 |
| Target tag | 1 |
| Channel frequency | 915MHz |

| | |
|---|---|
| Fading | No |
| Inference of Interchannel | No |
| Data rate | 2 Mbps |
| SNR based signal reception | 10 |
| Transmission power of RFID | -45 dBm |
| Read range | 1.62 m |
| Range of sensing | 5.4 m |
| Range of inference | 7.1m |
| Control channel frequency | 930 MHz |
| Number of nodes | 50 |
| Sensitivity reader power | -70dBm |
| Radio Rx Sensitivity | -91 dBm |
| Transmission range | 30 m |
| Sensitivity tag power | -17dBm |

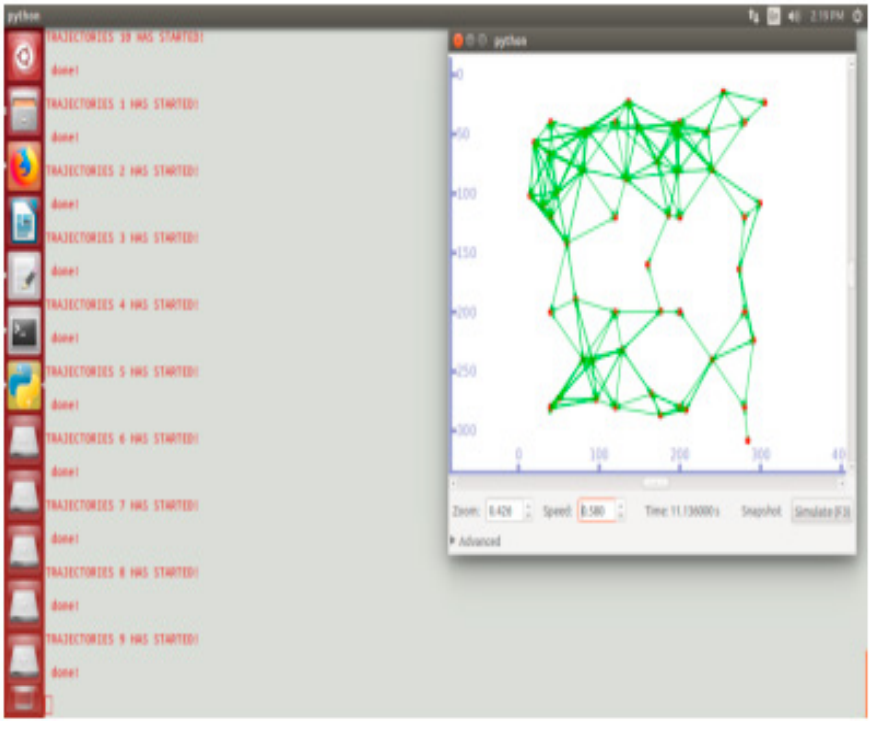



Figure 4 tracking process of target tag using proposed ASO-LSTM algorithm

Tracking process of target tag using proposed ASO-LSTM algorithm is shown in figure 4.

*b. Performance and comparative estimation*

The mean error and execution time for RSSI data is estimated and the outcomes are compared with traditional methods to validate the efficiency of proposed model with other conventional approaches.

Table 2 Comparative estimation of mean error rate and execution time (RSSI) [20]

| Technique | Mean error (m) | Std. dev. (m) | Mean time (ms) |
|---|---|---|---|
| ANN+ASO-LSTM | 0.973 | 0.509 | 28.2 |
| CNN | 1.130 | 0.537 | 27.4 |
| DNN | 1.224 | 0.563 | 17.6 |
| LANDMARC | 1.599 | 1.001 | 1.3 |
| WKNN | 1.626 | 1.014 | 1.3 |

Table 2 shows the mean error rate and execution analysis for the proposed ANN+ASO-LSTM localization and tracking of moving target tag with the values of existing models like CXNN, DNN, LANDMARC, and WKNN. The analysis reveals that the proposed system is offering better outcome than others that is error rate of proposed model is lower with effective mean time value. Comparison of mean error, standard deviation and mean time shown in figure 5.

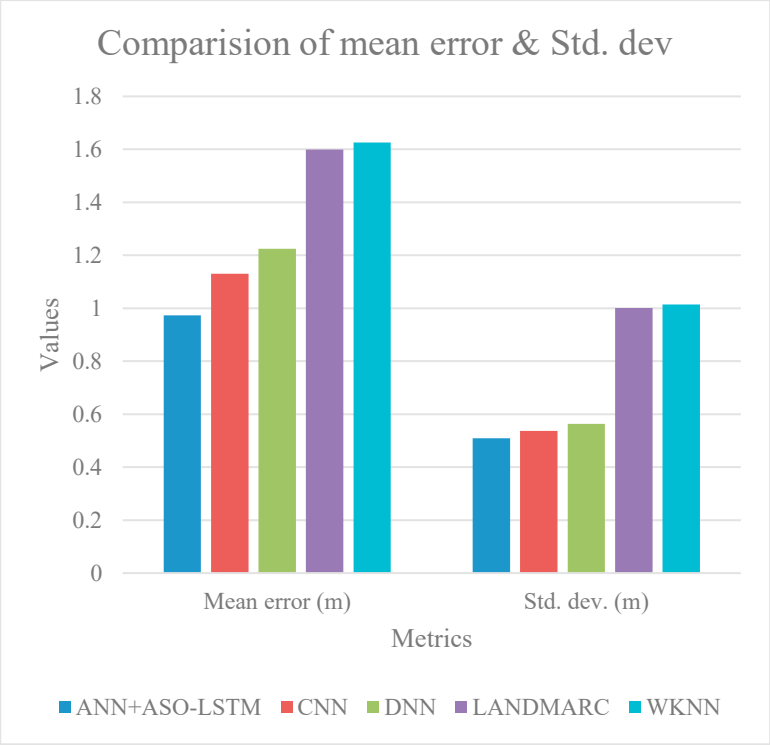
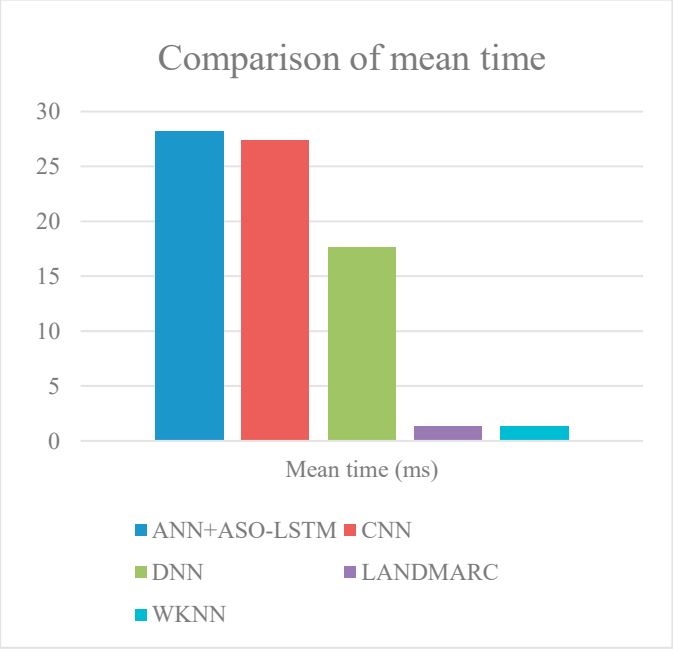



Figure 5 (a) Comparison of mean error and standard deviation




(b)

Figure 5 (b) Comparison of mean time

The tracking accuracy is employed for estimating the correctness or accurateness of proposed model. It is computed with respect to number of tags. The comparison of tracking accuracy in terms of proposed ASO-LSTM and existing model PSO-TRACK [20], MOIT [21] is given in table 3.

Table 3 Tracking accuracy comparison

| Technique | #of tags |
|---|---|
| Proposed ASO-LSTM | 98.99±0.5 |
| PSO-TRACK | 87.82±1.0 |
| MOIT | 80.6±1.5 |

Table 4 comparative analysis of SNR, # of tags, and # of trajectories

| Technique | #of tags | SNR | # of trajectories |
|---|---|---|---|
| Proposed ASO-LSTM | 98.99±0.5 | 4.7±0.5 | 4.11±0.05 |
| PSO-TRACK | 87.82±1.0 | 9.78±0.10 | 8.52±0.10 |
| MOIT | 80.6±1.5 | 18.38±0.15 | 22.42±0.15 |

Table 4 is the numerical estimation of tracking error which indicates that the tracking error average value with respect to SNR, number of tags, and number of trajectories. The comparison is made in terms of proposed ASO-LSTM and existing model PSO-TRACK [21], MOIT [22]. The analysis reveals that the proposed technique offers low tracking error on comparing traditional models. Thus, the model is effective than others.

## 5. Conclusion

This research paper focuses on achieving optimal localization and tracking of moving targets through the integration of deep learning and artificial intelligence models, organized into four distinct modules. The initial phase involves data collection and the estimation of Received Signal Strength Indicator (RSSI), providing essential insights into target movement and signal characteristics.

Subsequently, the Sequential Minimal Optimization (SMO) algorithm is employed to optimize the allocation of Virtual Reference Tags (VRT), enhancing the overall efficiency of the tracking system. The third module employs Artificial Neural Networks (ANN) for precise localization of unknown tags, leveraging the capabilities of machine learning to discern intricate patterns from the collected data.

Moving on, the fourth module utilizes the Aquila Swarm Optimization-Long short-term memory (ASO-LSTM) approach for tracking moving targets. This unique combination of swarm optimization and Long Short-Term Memory (LSTM) proves effective in dynamic tracking scenarios.

The final step entails estimating the target's location using the proposed technique, simulated within the NS3.26 network simulator. Performance evaluation is conducted using metrics such as Signal-to-Noise Ratio (SNR), tracking error, mean error, tracking accuracy, mean time, and standard deviation, offering a comprehensive assessment of the model's effectiveness.

To underscore the innovation of the proposed model, a comparative analysis is performed against traditional models. This comparative study serves to demonstrate the superior performance of the ASO-LSTM approach across the specified metrics.

In conclusion, this research endeavors to make a substantial contribution to the field by introducing a novel methodology that integrates deep learning, artificial intelligence, and optimization techniques for the optimal localization and tracking of moving targets. The sophistication of the approach is expected to advance the current understanding and capabilities in target tracking systems. Detailed results, discussions, and considerations for limitations and future work are provided throughout the paper to offer a thorough exploration of the research findings.

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